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1. (a) Show that, for $r > 0$,

$$\ln \left(1 + \frac{2}{r(r+3)} \right) = \ln \left(\frac{r+1}{r} \right) - \ln \left(\frac{r+3}{r+2} \right) \quad [1]$$

(b) Hence, using the method of differences, show that for integer $n \geq 1$,

$$\sum_{r=1}^n \ln \left(1 + \frac{2}{r(r+3)} \right) = \ln \left(\frac{3(n+1)}{n+3} \right) \quad [4]$$

2. Let a be a positive constant.

(a) Use the method of differences to find

$$\sum_{r=1}^n \frac{1}{(r+a)(r+a+1)(r+a+2)}$$

in terms of n and a .

[4]

(b) Hence find the value of a for which

$$\sum_{r=1}^{\infty} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{24}$$

[3]

3. (a) Show that, for $r \geq 2$,

$$\frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = \sqrt{r(r+2)} - \sqrt{r(r-2)} \quad [2]$$

- (b) Hence use the method of differences to determine

$$\sum_{r=2}^n \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}}$$

giving your answer in simplest form.

[3]

- (c) Hence show that

$$\sum_{r=3}^7 \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = A\sqrt{2} + B\sqrt{3} + C\sqrt{7}$$

where A , B and C are integers to be determined.

[2]

4. Use the method of differences to show that

$$\sum_{r=1}^n \frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{n(an+b)}{(2n+1)(2n+3)}$$

where a and b are integers to be determined.

[6]

5. (a) Use the method of differences to prove that for $n \geq 1$

$$\sum_{r=1}^n \ln \left(1 + \frac{2}{r(r+3)} \right) = \ln \left(\frac{3(n+1)}{n+3} \right) \quad [4]$$

- (b) Hence find the exact value of

$$\sum_{r=25}^{74} \ln \left(\left(1 + \frac{2}{r(r+3)} \right)^{28} \right)$$

Give your answer in the form $a \ln \left(\frac{b}{c} \right)$ where a , b and c are integers to be determined. [3]

6. Using partial fractions and a method of differences, prove that for positive integer n ,

$$\sum_{r=1}^n \frac{6}{r(r+1)(r+2)} = \frac{n(an+b)}{2(n+1)(n+2)}$$

where a and b are constants to be found.

[6]

7. Let $T_n = (n+1)^3 + (n+2)^3 + \cdots + (2n)^3$.

(a) By considering $(r+1)^4 - (r-1)^4$, use the method of differences to prove that

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2 \quad [5]$$

(b) Show that

$$T_n = \frac{1}{4}n^2(an^2 + bn + c)$$

where a , b and c are integers to be determined. [3]

(c) State the value of

$$\lim_{n \rightarrow \infty} \frac{T_n}{n^4}$$

[1]

8. (a) By expressing $\frac{1}{r(r+2)}$ in partial fractions, show that

$$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)} \quad [5]$$

- (b) Hence determine the value of

$$\sum_{r=1}^{\infty} \frac{1}{r(r+2)} \quad [2]$$

9. Show that

$$\frac{1}{1 \times 4} + \frac{1}{2 \times 5} + \frac{1}{3 \times 6} + \cdots + \frac{1}{n(n+3)} = \frac{11}{18} - \frac{an^2 + bn + c}{3(n+1)(n+2)(n+3)}$$

where a , b and c are constants to be found.

[6]

10. (a) Use standard results from the formula booklet to show that

$$\sum_{r=1}^N r(2r-1)(2r+1) = \frac{1}{2}N(N+1)(2N^2+2N-1) \quad [3]$$

(b) Express $\frac{6r+13}{(2r-1)(2r+3)}$ in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^N \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1}$$

in terms of N .

[4]

(c) Deduce the value of

$$\sum_{r=1}^{\infty} \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1} \quad [1]$$

11. (a) Show that

$$(2n)^5 - (2n-2)^5 \equiv 10(2n-1)^4 + 20(2n-1)^2 + 2 \quad [2]$$

(b) Hence, using the method of differences, show that for all positive integer values of n ,

$$\sum_{r=1}^n (2r-1)^4 = \frac{1}{15}n(2n-1)(2n+1)(an^2 + bn + c)$$

where a , b and c are integers to be determined.

[7]

12. Let

$$S_n = \sum_{r=1}^n \frac{1}{(3r-2)(3r+4)}$$

where $n \geq 1$.

(a) Use the method of differences to show that

$$S_n = \frac{n(15n+17)}{8(3n+1)(3n+4)} \quad [6]$$

(b) Show that, for any number k greater than $\frac{24}{5}$, if the difference between $\frac{5}{24}$ and S_n is less than $\frac{1}{k}$, then

$$n > \frac{k-15+\sqrt{k^2+81}}{18} \quad [6]$$